Harnessing Patient-generated data to Enhance Knowledge of Rare Diseases: A Feasibility Study

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Learning Objectives

- Explain the **value and characteristics of patient-generated data** in rare disease research
- **Compare and contrast** patient-reported rare disease symptoms with clinical literature
- Explore how patient-generated data from social media has the **potential to improve the early detection of rare diseases**

Background

- **>7000 rare diseases globally**, often **underrepresented** in clinical literature and difficult to diagnose
- Diagnostic delay common: **average of 5-7 years** from first symptom onset → rare disease diagnosis
- New data source: **social media**, Facebook groups
- Prior lit: Rare disease social media video analysis (Park, 2025)
- Patients often share **rich symptom & comorbidity info** within their **rare disease narratives**
- Narratives capture **real-world experiences** and first-person perspectives of rare disease patients
- Immense potential to enhance rare disease knowledge

Project Goals

1) Characterize rare disease patient narratives, eg. **frequently reported symptoms/comorbidities** from patient online posts
2) Compare patient-reported symptoms/comorbidities with **clinical literature** (PubMed)
3) Explore how insights from social media and patients lived-experiences have the potential to **improve the detection of rare diseases**

Methods

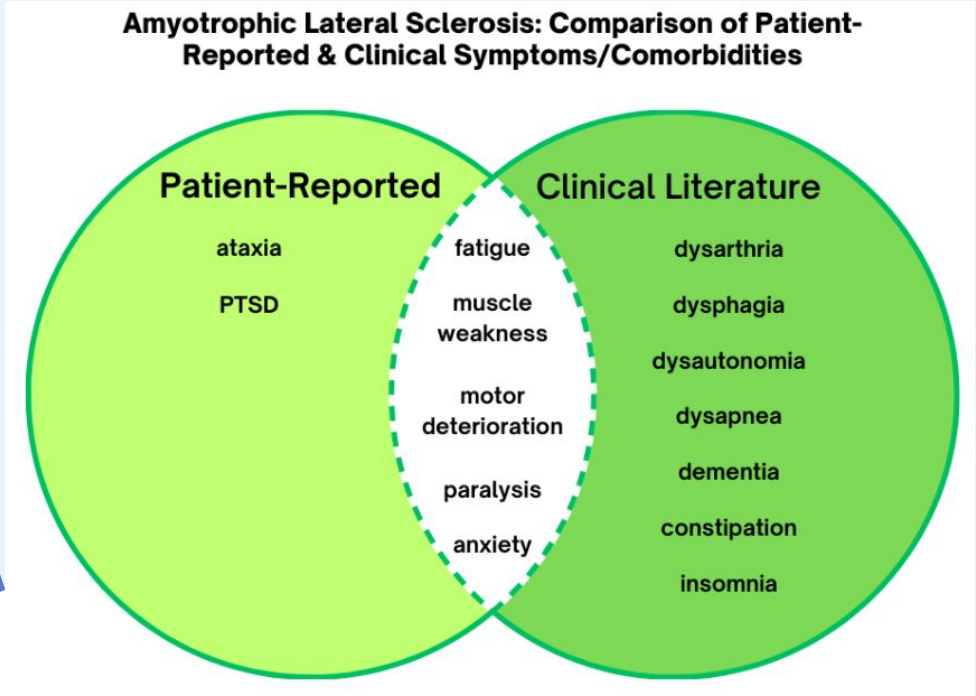
- **249** patient's narratives from 2019-2025 → **127 rare diseases**
- Source: Facebook "Rare Disease Warriors" group
- **Characterization** of patient data (using GPT-4 with manual verification)
  - Symptoms/comorbidities
  - Country
  - Rare disease diagnosis
  - Age at diagnosis, etc.
- **Standardization** of rare diseases using **ICD-10 mappings and NORD database;** symptoms/comorbidities using **HPO database**
- **Literature review** of clinical characteristics of **top 3 most common rare diseases**
- **Comparison** of patient-reported symptoms with PubMed clinical symptoms

Results - Patient Narrative Characterization

- **48.6%** of patient posts were from the US
- **Most common age groups for diagnosis:** young adults, adults, infants
- **Most frequent symptoms:** Motor impairment, ataxia, pain, fatigue
- **Most common disorders:** Spinocerebellar ataxia, Ehlers-Danlos Syndrome, Amyotrophic Lateral Sclerosis → neurological
- **Most commonly mentioned narrative themes:** psychosocial impact and coping and resilience

Number of Patients		249
Number of Rare Diseases		127
Gender	**Female**	161
	Male	105
Undiagnosed		12

Results - Comparison of Symptoms



Rare Disease	Jaccard index
General SCA	0.258
SCA2	0.158
SCA14	0.250
Friedreich's ataxia	0.211
General EDS	0.192
Hypermobile EDS	0.194
General ALS	0.152

Discussion & Limitations

- High rates of motor impairment & ataxia → **predominance of neurological disorders**
- Fatigue and pain → common across **many chronic illnesses** and patient-reported burdens
- Narratives focus on **quality-of-life** and **functional impacts**, often missing in literature
- Limitations: **Small sample size**, manual data collection, incomplete data, sampling bias, recall-biased patient data

Conclusion

- Patient-generated data reveals **underreported symptoms**
- Patient perspectives **complement clinical datasets,** revealing **quality-of-life impacts** often overlooked in diagnostics
- Strong potential to **improve early rare disease recognition**, guide research priorities, and inform patient-centered care
- Future Directions:
  - Collect more data (**larger sample size)**
  - Explore **LLMs/AI** to enhance rare disease detection, patient support
  - **Integrate** patient-generated data with clinical research

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